

13 **Ans: B**

Given: **Actual** gas pressure = pressure **gauge reading** (of 200 kPa) + **atmospheric pressure** (of 100 kPa)

Thus initial actual gas pressure $p_i = 200 + 100 \text{ kPa} = 300 \text{ kPa}$, at initial temp of 25°C

Assuming vol of tyre = const, actual gas pressure $p \propto T$. Given $T_f = 50^\circ\text{C}$

→ Final actual gas pressure $p_f = (T_f/T_i) \times p_i = (50 + 273.15)/(25 + 273.15) \times 300 \text{ kPa}$
 $= 325 \text{ kPa}$

Thus final pressure **gauge** reading = $325 - 100 = 225 \text{ kPa}$

Thus $\Delta p/p_i$ in the pressure **gauge** readings = $(225 - 200)/200 = 0.125$

In terms of percentage, $= 0.125 \times 100\% = 12.5 \%$

14 **Ans: C**